

Riding out of Tillage: The Gig Economy and Rural Factor Reallocation in China

By HUANXI ZHANG*

This paper studies how digital platform entry affects labor allocation, land transfer, and agricultural outcomes in rural China. Using newly constructed city-level platform entry dates matched to household panel data from the National Fixed Point Survey, 2009–2017, we exploit the staggered rollout of major gig platforms in a difference-in-differences design. Platform entry reduces household agricultural participation and increases the probability of land transfer-out by 2.2 percentage points. These effects are stronger among younger and more educated individuals, while land fragmentation constrains land reallocation. On the agricultural side, platform entry lowers food production overall. Among land-transferring-in households, it increases capital input per mu and crop revenue per mu while reducing food yield per mu; among non-transferee households, it lowers crop revenue per mu. Overall, the findings suggest that gig-platform expansion raises the opportunity cost of farming and accelerates reallocation out of agriculture, with uneven effects on agricultural performance. (JEL: J43, O13, O33, R23, Q15.)

Keywords: *Gig Economy, Land Transfer, Agricultural Transformation, Platform Economy*

I. Introduction

The widespread adoption of digital platforms and algorithmic matching technologies has altered traditional employment relationships.¹ Digital platforms and on-demand matching technologies have enabled workers to provide short-term, task-based services to consumers (Cramer and Krueger 2016). This shift away from traditional long-term employment toward more flexible, platform-mediated work is commonly referred to as the gig economy.²

Globally, such flexible employment has become an increasingly important component of labor-market adjustment (Abraham et al. 2019; Vertesi et al. 2020; Wang, Gao and Zhang 2025). Relative to the more gradual expansion observed in advanced economies, the gig economy in China has grown rapidly over the past decade. Accord-

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¹See *OECD* https://www.oecd.org/en/publications/algorithmic-management-in-the-workplace_287c13c4-en.html

²See the *World Economic Forum report* <https://www.weforum.org/stories/2024/11/what-gig-economy-workers/>.

ing to recent estimates, the number of gig workers in China has reached approximately 200 million, accounting for around 40 percent of the urban workforce.³

Existing work on the gig economy has focused primarily on urban areas and the organization of urban labor markets. This literature highlights how platforms generate employment opportunities and reduce information frictions in labor markets (Cramer and Krueger 2016; Chen et al. 2019), while also raising concerns regarding job insecurity and algorithmic control (De Stefano 2016; Berger et al. 2019). Recent work also studies the broader consequences of platform expansion, showing that the gig economy influences firms’ capital-labor substitution and entrepreneurial decisions (Burtch, Carnahan and Greenwood 2018; Barrios, Hochberg and Yi 2022; Denes, Lagaras and Tsoutsoura 2025). However, there is little empirical evidence on how platform expansion affects the agricultural sector, especially in developing economies with pronounced rural-urban segmentation.

Development economics has long relied on classic dual-economy models (Lewis 1954; Harris and Todaro 1970) to understand the reallocation of labor out of agriculture and the broader process of structural transformation. A central implication of these models is that cross-sectoral factor reallocation relies heavily on cross-regional out-migration. Since 2012, however, the acceleration of rural land transfer has coincided with a slowdown in interregional migration (Figure 1), suggesting that land reallocation may be becoming increasingly decoupled from traditional migration.

Relative to standard formal employment, platform-based work may be particularly accessible to rural workers, as it typically requires limited formal skills, has low entry barriers, and offers relatively attractive short-run returns.³ Such flexible nonfarm alternatives raise the opportunity cost of farming. Descriptive evidence from the China Labor-force Dynamics Survey (CLDS) is consistent with this interpretation: as shown in Figure 2a, roughly 80 percent of gig workers in the CLDS sample hold an agricultural *hukou*, and a substantial proportion previously engaged in agricultural production before entering gig work (Figure 2b).⁴ As nonfarm opportunities become more attractive, rural households may reallocate labor within the household and further adjust by transferring out land-use rights.

These aggregate patterns and descriptive facts motivate two questions. First, does the expansion of the gig economy reallocate rural household labor away from agriculture? Second, does it facilitate the reallocation of land-use rights through rental markets? To address these questions, we construct city-level platform entry dates from Wayback Machine⁵ and cross-validate them with Zhilian Recruitment data to construct city-level entry dates for five major platform firms—Meituan, Ele.me,

³See *The Economist* <https://www.economist.com/leaders/2025/09/18/chinas-200m-gig-workers-are-a-warning-for-the-world>.

⁴The CLDS identifies gig workers using the category of “unfixed employment,” which includes odd-job workers, street vendors, independent domestic workers, self-employed drivers, and artisans. This classification is broader than our focus on platform-based gig work. To assess whether it nonetheless provides a reasonable proxy, we extract respondents’ self-reported job descriptions and construct a word cloud (Figure A1). The most frequent terms are closely related to occupations commonly associated with contemporary platform work, supporting the relevance of this measure for our descriptive analysis.

⁵The Wayback Machine (<https://web.archive.org/>) is a digital archive of the World Wide Web founded by the Internet Archive, allowing users to go “back in time” to see how websites looked in the past.

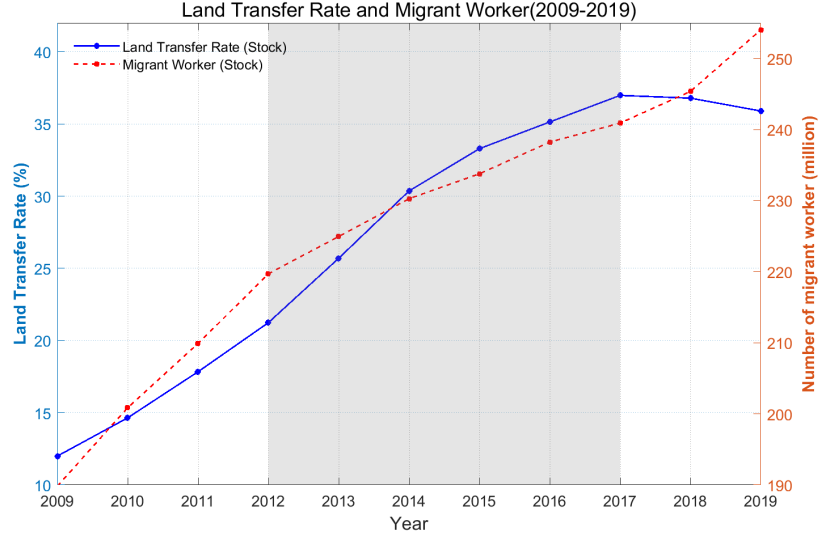
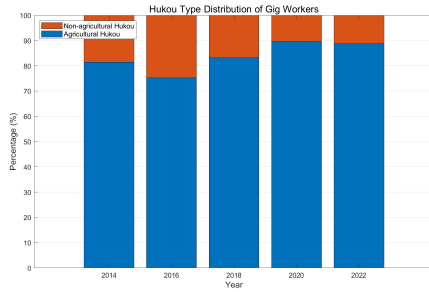
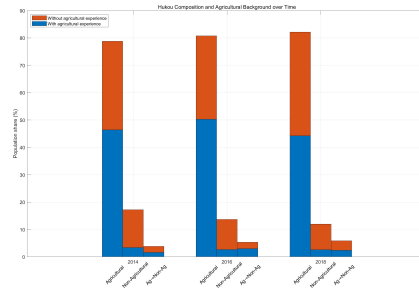


Figure 1. National Trends in Land Transfer and Population Mobility, 2009–2019

Notes: This figure plots the joint evolution of the rural land transfer rate and the aggregate stock of migrant labor. The data are drawn from the National Bureau of Statistics (NBS) and the Ministry of Agriculture and Rural Affairs (MARA). Both series are measured as end-of-period stocks rather than annual flows. The land transfer rate is defined as the cumulative area of leased household-contracted farmland divided by total household-contracted arable land. The migrant labor measure captures the total stock of rural-to-urban migrant workers. The divergence between the two series in the latter part of the sample is consistent with a weakening association between land consolidation and labor out-migration through the traditional migration channel.



(a) Hukou type distribution



(b) Hukou composition and prior agricultural experience

Figure 2. Hukou Composition and Prior Agricultural Experience of Gig Workers

Notes: This figure plots (a) the shares of gig workers with rural (agricultural) *hukou* and non-rural *hukou*, and (b) *hukou* composition and prior agricultural production experience before entering gig work. Data are sourced from the China Labor-force Dynamics Survey (CLDS). Percentages are computed using survey weights.

Koubei, Baidu Waimai, and 58.com. We then match these entry dates to household panel data from the National Fixed Point (NFP) Survey for 2009–2017 and estimate a staggered difference-in-differences model.

Our results show that platform entry affects rural factor allocation in several ways. First, platform entry reduces household agricultural participation and increases land transfer-out on both the extensive and intensive margins. It raises the probability of land transfer-out by 2.2 percentage points and increases the transferred-out area by approximately 0.032 log points. These responses are heterogeneous across households and individuals. The decline in agricultural participation is more pronounced among younger and more educated individuals, while land fragmentation significantly constrains land reallocation. Platform entry also has important consequences for agricultural outcomes. In the full sample, it lowers food production. Among land-transferring-in households, it increases capital input per mu and crop revenue per mu while reducing food yield per mu, whereas among non-transferee households it lowers crop revenue per mu. Taken together, these findings suggest that gig-platform expansion raises the opportunity cost of farming and accelerates reallocation out of agriculture, with uneven effects on agricultural performance.

Our paper contributes to the literature on the gig economy by extending the analysis from urban labor markets to rural households and agricultural factor markets. Whereas existing work has focused primarily on urban employment, wages, and labor relations (Cramer and Krueger 2016; Chen et al. 2019; Barrios, Hochberg and Yi 2022; Shin et al. 2023; Denes, Lagaras and Tsoutsoura 2025), much less is known about how platform expansion affects labor allocation and land use in rural areas. Using household-level data from rural China, we show that digital platforms affect both agricultural participation and land transfer.

More broadly, the paper contributes to the literature on rural structural transformation by showing that factor reallocation can occur even in the absence of large-scale migration. A large literature links changes in agricultural labor allocation and land use to rural out-migration and urbanization, emphasizing spatial mobility as a central driver of structural change (Kuznets 1957; Taylor and Martin 2001; Rozelle, Taylor and DeBrauw 1999; Michaels, Rauch and Redding 2012; Cao and Birchenall 2013). In contrast, our results point to a more localized process of adjustment. They suggest that expanded nonfarm opportunities can induce households to reduce agricultural participation and reallocate land through rental markets, even without large-scale spatial migration.

The remainder of the paper is organized as follows. Section 2 presents background information on China’s rural land market and the expansion of the gig economy. Section 3 describes the data and empirical strategy. Section 4 presents the main results on agricultural labor participation and land transfer. Section 5 examines heterogeneity in these effects. Section 6 provides further analysis on the effects of gig-economy expansion on employment, income, and agricultural output. Section 7 concludes.

II. Background

A. *The Rural Land Market in China*

Land rental markets play an important role in the allocation of agricultural resources. In many developing economies, weak property-rights protection and high transaction costs impede the transfer of land from less productive to more productive operators, contributing to persistent land misallocation (Adamopoulos and Restuccia 2014).

China’s rural land system is rooted in the Household Responsibility System. Under this system, village collectives retain land ownership, while rural households hold contract and management rights, with initial allocations based largely on household size (Lin 1988, 1992). Although the legal framework has allowed transfers of contracted land since 1988, the rural land rental market remained limited before the 2000s. In the absence of stable nonfarm income opportunities, many households continued to cultivate their own land rather than transfer it to more productive operators, contributing to persistent inefficiencies in land allocation (Kung 2002).

As urbanization progressed and the nonfarm sector expanded, rising returns to nonfarm employment increased the opportunity cost of agricultural labor and encouraged labor reallocation out of agriculture (Harris and Todaro 1970; Kuznets 1957; Tian, Xia and Yang 2020). At the same time, stronger tenure security and lower transaction costs made land rental more feasible, allowing changes in labor allocation to translate more readily into land transfers (Benjamin and Brandt 2002; Jacoby, Li and Rozelle 2002; Jin and Deininger 2009). The Rural Land Contracting Law of 2003, later revised in 2018, strengthened the legal basis for transferring management rights while retaining household contract rights.

However, land transfers continue to face substantial frictions, including information asymmetries, localized market constraints, and high transaction costs embedded in traditional social networks. Consequently, the pace and scale of land consolidation in rural China remain well below the efficiency-maximizing level, accompanied by persistent land fragmentation. Accelerating land transfers and dismantling barriers to factor mobility have thus emerged as a central agenda linking agricultural productivity improvements with broader rural structural transformation.

B. *The Gig Economy*

In this paper, the gig economy refers to platform-mediated, on-demand work in which digital platforms match workers to short-term service tasks. Relative to standard formal employment, this type of work typically involves lower entry barriers, greater schedule flexibility, and more frequent payment. Consequently, platform work is particularly effective at absorbing marginalized laborers transitioning between the agricultural and nonfarm sectors. Crucially, relying heavily on localized consumption markets and fulfillment networks, platform expansion predominantly stimulates local nonfarm employment rather than inducing traditional long-distance spatial migration. For rural households, the proliferation of platform-based opportunities fundamentally alters intra-household labor allocation by simultaneously expanding local

nonfarm employment prospects and elevating the opportunity cost of agricultural labor.

The expansion of smartphone use, mobile payment systems, and digital platform services contributed to the rapid growth of gig work in China during the mid-2010s. Food delivery, ride-hailing, and other local service platforms became increasingly important sources of flexible employment during this period. The platforms in our sample were among the major firms operating in these segments. Characterized by high-frequency demand, highly standardized service protocols, and strong complementarities with location-based matching technologies, these sectors rapidly penetrated urban markets.

Platform expansion also displayed substantial geographic variation, with entry occurring earlier in large coastal cities and later in inland and lower-tier cities. Figure 3 illustrates this staggered diffusion between 2014 and 2017. We use this variation in entry timing to construct a city-level indicator of gig-platform entry in the empirical analysis.

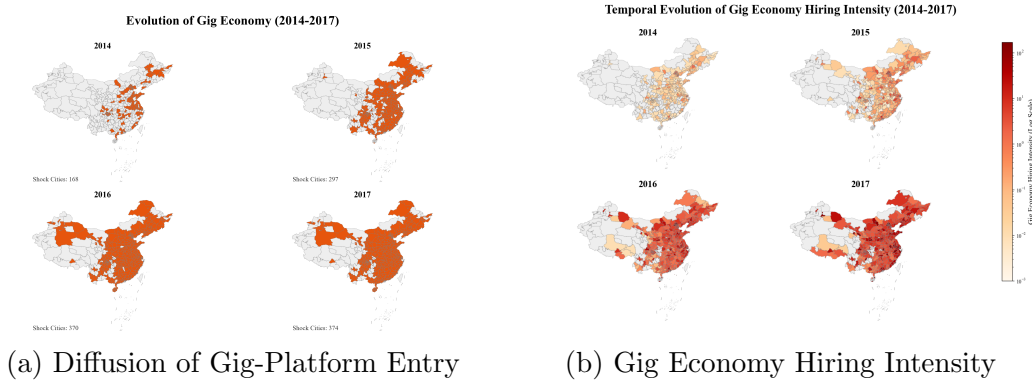


Figure 3. Spatial and Temporal Evolution of the Gig Economy in China

Notes: This figure illustrates the spatial and temporal expansion of the gig economy across Chinese cities. To ensure the reliability of the identified treatment timing, we determine the initial entry of major gig platforms into each municipality using historical web archives from the Wayback Machine, cross-validated against data from Zhilian Recruitment. Panel (a) plots the staggered entry path of major platform firms at the city level. Panel (b) displays the gig economy recruitment intensity index, defined as the natural logarithm of the ratio of gig vacancies to the total municipal population plus one, expressed as $\ln(\text{Gig Postings}/\text{Population} + 1)$.

III. Data and Empirical Strategy

A. Data Sources

Our primary data source is the National Fixed Point (NFP) Survey, conducted by the Research Center for Rural Economy (RCRE). The NFP is a nationally representative village-household-individual panel covering approximately 300 villages and

20,000 households across 31 provinces, with detailed information on agricultural production, land use, and labor allocation. We use the publicly available waves, including the full sample from 2009 to 2015 and a 30 percent random subsample from 2016 to 2017. Because land transfer and labor allocation are household decisions in rural China, our baseline analysis is at the household level. We aggregate individual employment outcomes to the household and define our main outcome as an indicator equal to one if any household member engages in agricultural production. In heterogeneity analyses, we use the underlying individual-level data to examine differential responses by age and education. All continuous variables are winsorized at the 2.5th and 97.5th percentiles.

Our key explanatory variable is the earliest timing of gig-platform entry in each prefecture-level city. We construct this measure using historical web archives from the Wayback Machine for five major platforms—Meituan, Ele.me, Koubei, Baidu Waimai, and 58.com—collectively accounting for the dominant share of food-delivery and ride-hailing activity in China over the sample period. For each platform-city pair, we identify the earliest archived date on which the name of a prefecture-level city appears on the platform’s webpage.⁶ We then define city-level entry as the earliest such date observed across platforms.⁷

Because historical web archives may be incomplete and website launches may not perfectly coincide with local platform operations, we cross-validate the city-level entry measure using external recruitment data. Specifically, we compare Meituan’s archived entry dates with the earliest recruitment records for gig-related occupations—including delivery riders, ride-hailing drivers, and domestic workers—from the Zhilian Recruitment database. When the two sources differ, we use the earlier date recorded in the recruitment data as the city’s entry year.⁸

To assess whether the treatment variable captures the expansion of the gig economy in a meaningful way, we estimate an event-study specification using local gig recruitment intensity as the outcome. The underlying city-year panel covers 365 cities, while the event-study sample includes 297 prefecture-level cities over 2009–2017 because of missing values in control variables, for a total of 2,673 city-year observations. The specification includes city and year fixed effects, along with the same time-varying city-level controls used in the baseline analysis.^{9,10} Figure 4 reports the dynamic estimates. The coefficients in the pre-entry periods are close to zero and statistically insignificant, providing little evidence of differential pre-trends. By contrast, the estimates rise sharply upon platform entry and remain positive thereafter. This pattern is consistent with the treatment variable capturing the timing of platform expansion.

⁶Because archived web records are frequent and voluminous, we sample archived pages at weekly intervals when reconstructing platform entry histories. Since our object of interest is the year of city-level entry rather than the precise date, this procedure improves computational tractability without materially affecting the timing measure.

⁷Meituan provides the most complete archive coverage over the sample period. Other platforms increasingly shifted toward mobile-based operations in later years, leaving fewer traceable web records.

⁸Appendix Figure A2 reports the distribution of Meituan delivery entry years and the final city-level entry years used in the analysis.

⁹The controls include the logarithm of population, the logarithm of GDP per capita, the tertiary sector’s share of GDP, the urbanization rate, and the mobile-phone penetration rate.

¹⁰Gig recruitment intensity is defined as $\ln(\text{Gig Postings}/\text{Population} + 1)$.

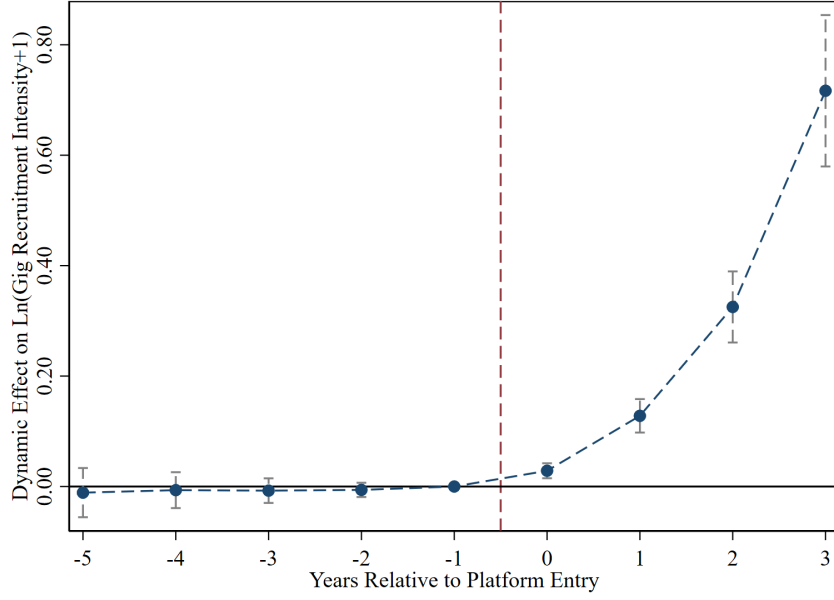


Figure 4. The Dynamic Impact of Gig-Platform Entry on Local Recruitment Intensity

Notes: This figure plots the estimated coefficients and their corresponding 95 percent confidence intervals from an event-study specification. The dependent variable is the logarithm of gig recruitment intensity. The event window spans five years prior to platform entry and three years after, with the year immediately preceding entry ($k = -1$) serving as the omitted baseline category. The regression controls for a vector of time-varying city characteristics, including log population size, log GDP per capita, the tertiary industry's share of GDP, urbanization rate, and mobile phone penetration. All specifications incorporate city and year fixed effects. Standard errors are clustered at the city level. The sample consists of a balanced panel of 297 Chinese cities from 2009 to 2017.

B. Variable Definitions

Outcome Variables. Our main outcomes are measured at the household level. We first examine an indicator for household agricultural participation, equal to one if any household member engages in agricultural production in a given year. We then study land-transfer outcomes along both the extensive and intensive margins. The extensive-margin outcome is an indicator equal to one if the household transfers out any land during the year. The intensive-margin outcomes are the log land area transferred out and the log total transfer area, defined as the sum of land transferred in and land transferred out. In the further analysis, we also examine household income and employment allocation, including total household income, local work participation, local nonfarm wage income, out-migration work participation, and migrant nonfarm wage income. We additionally examine agricultural outcomes, including food production, food yield per mu, crop revenue per mu, food revenue per mu, cash revenue per mu, and productive fixed assets per mu. Productive fixed assets per mu are measured as the stock of agricultural machinery and equipment divided by sown area. All continuous variables are transformed using $\ln(x + 1)$ and winsorized at the 2.5th and 97.5th percentiles.

Treatment Variable. The treatment variable is a city-year indicator for gig-platform entry. For each prefecture-level city, it equals one in the year of the earliest identified platform entry and in all subsequent years, and zero otherwise. The data construction and validation procedure for this variable is described in Section 3.1.

Control Variables. All baseline specifications include time-varying prefecture-level controls: the logarithm of population size, the logarithm of GDP per capita, the tertiary sector’s share of GDP, the urbanization rate, and mobile-phone penetration.

C. Empirical Strategy

To identify the causal effects of gig-platform entry on rural households, we exploit the staggered timing of platform entry across prefecture-level cities. We estimate the following baseline difference-in-differences specification:

$$(1) \quad Y_{ivct} = \alpha + \beta \text{Entry}_{ct} + X'_{ct}\gamma + \delta_i + \delta_t + \lambda_v \times t + \epsilon_{ivct}$$

where Y_{ivct} denotes the outcome for household i in village v , prefecture-level city c , and year t ; Entry_{ct} is an indicator equal to one if city c has had a platform enter by year t and zero otherwise; X'_{ct} is a vector of time-varying city-level controls; δ_i denotes household fixed effects; δ_t denotes year fixed effects; and $\lambda_v \times t$ denotes village-specific linear time trends that absorb gradual village-level changes. Standard errors are clustered at the prefecture-level city level, the level of treatment assignment.

To examine dynamic treatment effects and assess pre-trends, we estimate the following event-study specification. Let E_c denote the year of first platform entry in city c , and define the event-time indicator $D_{ct}^k = \mathbb{I}(t - E_c = k)$. Using the year before entry ($k = -1$) as the omitted category and restricting the event window to $k \in [-5, 3]$, we

Table 1—Summary Statistics

Variables	Source	Obs.	Mean	Std. Dev.	Min	Max
<i>Panel A: Explanatory Variable</i>						
Gig Platform Entry	1	142,545	0.242	0.428	0.000	1.000
<i>Panel B: Core Outcome Variables</i>						
Household Agricultural Participation	2	154,378	0.906	0.292	0.000	1.000
Land Transfer-Out Participation	2	154,378	0.126	0.332	0.000	1.000
Log(Transfer-Out Area+1)	2	154,378	0.189	0.577	0.000	6.516
Log(Total Transfer Area+1)	2	154,378	0.270	0.709	0.000	7.013
<i>Panel C: Other Outcome Variables</i>						
Local Work Share	2	132,584	0.657	0.368	0.000	1.000
Out-Migration Work Share	2	141,871	0.377	0.388	0.000	1.000
Log(Total Household Income+1)	2	154,378	10.547	1.026	0.000	16.811
Log(Local Nonfarm Wage Income+1)	2	154,378	3.486	4.621	0.000	14.403
Log(Migrant Nonfarm Wage Income+1)	2	154,377	4.345	4.989	0.000	14.039
Log(Food Production+1)	2	154,378	4.682	3.907	0.000	15.425
Log(Food Yield per Mu+1)	2	149,707	2.499	2.988	0.000	6.644
Log(Crop Revenue per Mu+1)	2	154,378	5.272	3.393	0.000	9.250
Log(Food Revenue per Mu+1)	2	154,377	4.061	3.339	0.000	7.513
Log(Cash Revenue per Mu+1)	2	154,378	3.164	3.809	0.000	9.554
Log(Capital per Mu+1)	2	154,378	4.281	3.413	0.000	9.657
<i>Panel D: Individual-Level Variables</i>						
Individual Agricultural Participation	2	408,488	0.871	0.335	0.000	1.000
Age	2	531,086	40.918	18.888	0.100	99.000
Years of Schooling	2	531,086	7.189	3.352	0.000	25.000
<i>Panel E: City-Level Controls</i>						
Ln Population Size	3	2,700	5.792	0.902	-3.219	8.129
Ln GDP per Capita	3	2,700	10.525	0.893	8.391	20.399
Industrial Structure	3	2,700	0.389	0.103	0.098	1.100
Urbanization Rate	3	2,700	0.381	0.201	0.015	1.000
Log Mobile Penetration	3	2,700	0.993	0.962	0.134	10.166

Source:

1. Wayback Machine and Zhilian Recruitment Data.
2. National Fixed Point (NFP) Survey, including the full sample from 2009 to 2015 and a 30 percent random subsample from 2016 to 2017.
3. China City Statistical Yearbook.

estimate:

$$(2) \quad Y_{ivct} = \alpha + \sum_{k \neq -1} \beta_k D_{ct}^k + X'_{ct} \gamma + \delta_i + \delta_t + \lambda_v \times t + \epsilon_{ivct}$$

The coefficients β_k trace the evolution of outcomes relative to the year before entry. We use the pre-entry coefficients ($k \leq -2$) to assess the parallel-trends assumption and the post-entry coefficients ($k \geq 0$) to characterize the timing and persistence of the effects of gig-platform entry. Standard errors are clustered at the prefecture-level city level.

IV. Empirical Results

Before turning to the formal analysis, we provide descriptive evidence on the relationship between gig-economy development and rural household behavior. Using the city-level gig recruitment intensity index, Figure 6 plots the association between gig recruitment intensity and our main outcomes: household agricultural participation and land transfer. Cities with greater gig recruitment intensity tend to exhibit lower rates of household agricultural participation and more active land-transfer markets.¹¹

A. Effect of Gig-Platform Entry on Agricultural Labor Participation

We begin by examining whether the expansion of the gig economy affects household participation in agricultural production. Table 2 reports estimates from Equation (1). In Column 1, gig-platform entry reduces household agricultural participation by 1.2 percentage points. The estimate is unchanged after adding time-varying city-level controls in Column 2. Relative to the sample mean, this corresponds to a modest but meaningful decline along the extensive margin of farm work. The results indicate that platform entry reduces household participation in agriculture.

To assess the identifying assumption and characterize the timing of the response, Figure 5 presents event-study estimates based on Equation (2). The coefficients in the pre-treatment periods are close to zero and statistically indistinguishable from zero, which is consistent with the parallel-trends assumption. After platform entry, the estimated coefficients turn negative and are statistically distinguishable from zero, providing event-study evidence of a causal decline in agricultural participation in the post-entry periods.

The dynamic pattern further suggests that the estimated effects are largest in the first two years following platform entry and attenuate thereafter, consistent with a bounded labor-reallocation process in which the pool of marginal workers most responsive to the gig-wage opportunity is gradually exhausted. As these more mobile workers exit agricultural production, the remaining agricultural labor force may become less responsive to the same shock. Taken together, the dynamic evidence

¹¹The gig recruitment intensity index may be endogenous to local household behavior because of reverse causality and omitted variables. We therefore do not use this measure in the baseline identification strategy and instead treat it as descriptive evidence to motivate the empirical analysis.

is consistent with the baseline difference-in-differences results and corroborates the identifying assumption of parallel pre-trends.

Table 2—Effect of Gig-Platform Entry on Agricultural Labor Participation

Variables	Household Agricultural Participation	
	(1)	(2)
Gig Platform Entry	−0.012*** (0.004)	−0.012*** (0.004)
City-level Controls	No	Yes
Household FE	Yes	Yes
Year FE	Yes	Yes
Village Linear Trends	Yes	Yes
Observations	141,936	141,936
Adj. R-squared	0.63	0.63

Notes: This table reports baseline estimates of the effect of gig-platform entry on household agricultural participation. The dependent variable is an indicator for household agricultural participation, defined as whether any household member engages in agricultural work. Column 1 excludes time-varying city-level controls, while Column 2 includes them. All specifications include household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B. From Labor Exit to Rural Land Transformation

BASELINE RESULTS

We next examine household land-transfer responses to gig-platform entry. Table 3 reports baseline estimates from Equation (1). Columns 1 and 2 use an indicator for land transfer-out. The coefficient is 0.022 in both specifications and statistically significant at the 1 percent level. Relative to a sample mean of 12.6 percent (Table 1, Panel B), this implies a 2.2-percentage-point increase, or a 17.5 percent rise over the mean. The similarity between Columns 1 and 2 indicates that the extensive-margin result is not sensitive to the inclusion of time-varying city-level controls.

Columns 3 and 4 use the log transfer-out area as the dependent variable. The coefficient is 0.032 in both columns and remains statistically significant after adding city-level controls. Columns 5 and 6 use the log total transfer area, defined as the sum of land transferred in and land transferred out. The coefficient is 0.040 in both specifications and is also statistically significant. The results therefore indicate that platform entry increases land transfer-out on both the extensive and intensive margins.

Figure 7 reports the corresponding event-study estimates. The coefficients in the pre-entry periods are close to zero and provide no evidence of differential pre-trends. After entry, the coefficients turn positive and remain positive thereafter, consistent with an increase in land transfer-out following platform entry. The dynamic estimates reinforce the baseline difference-in-differences results.

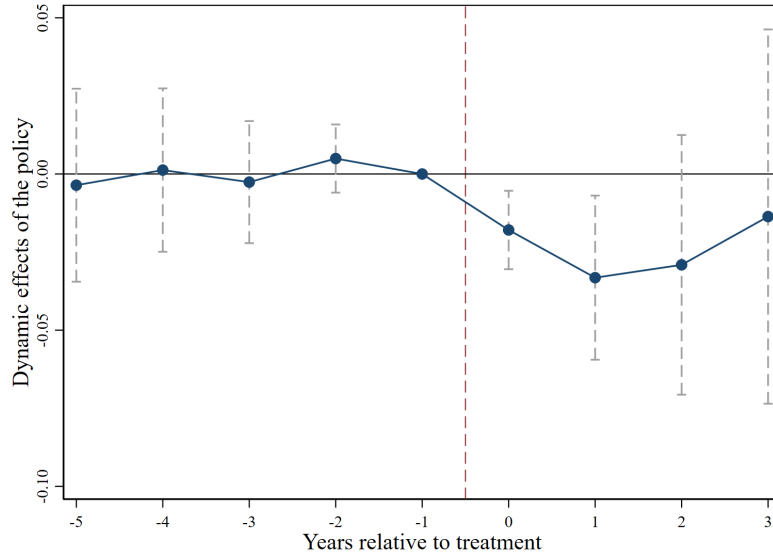


Figure 5. Dynamic Effects of Gig-Platform Entry: Agricultural Labor Participation

Notes: This figure plots event-study estimates of the effect of gig-platform entry on household agricultural participation. The markers and capped spikes represent ordinary least squares estimates and 95 percent confidence intervals based on standard errors clustered at the city level. The dependent variable is an indicator equal to one if any household member engages in agricultural work. The year immediately prior to platform entry is omitted as the reference period. All specifications incorporate time-varying city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends.

Table 3—Effect of Gig-Platform Entry on Land Transfer

Variables	Land Transfer-Out Participation		Log(Transfer-Out Area+1)		Log(Total Transfer Area+1)	
	(1)	(2)	(3)	(4)	(5)	(6)
Gig Platform Entry	0.022*** (0.008)	0.022*** (0.008)	0.032*** (0.012)	0.032*** (0.012)	0.040*** (0.013)	0.040*** (0.013)
City-level Controls	No	Yes	No	Yes	No	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes	Yes	Yes	Yes
Observations	141,936	141,936	141,936	141,936	141,936	141,936
Adj. R-squared	0.47	0.47	0.49	0.49	0.50	0.50

Notes: This table reports baseline estimates of the effects of gig-platform entry on rural land transfer. The dependent variables are an indicator for transfer-out participation, transfer-out area, and total transfer area, with the latter two measured as $\ln(x+1)$. Total transferred area includes both transfer-in and transfer-out land. All specifications include time-varying city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Gig Economy and Factor Reallocation

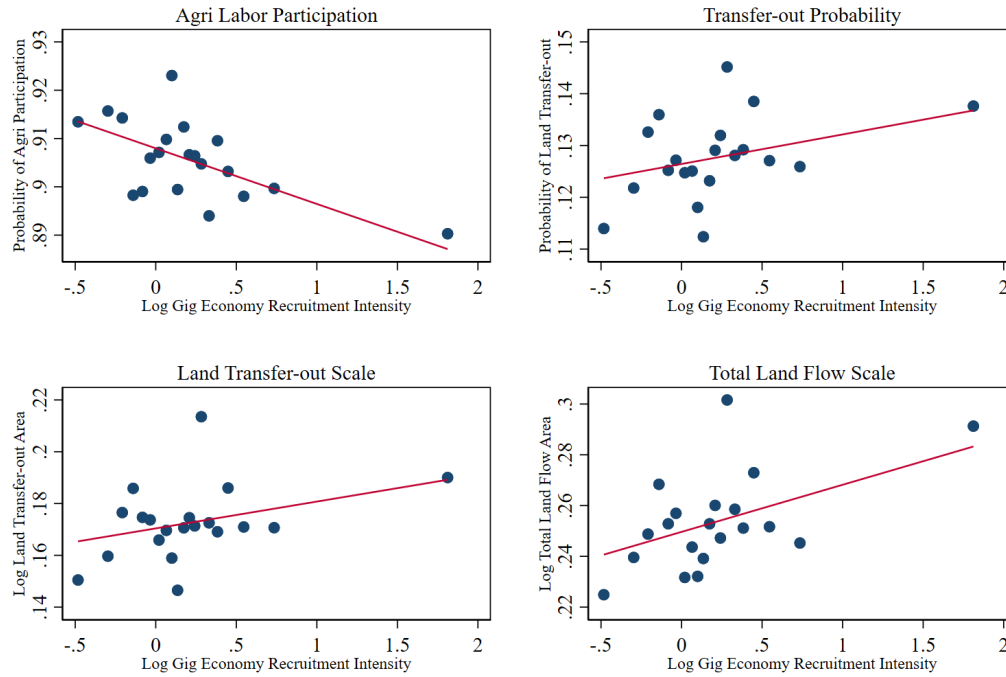


Figure 6. Correlations between Gig Intensity and Rural Household Outcomes

Notes: This figure plots bivariate relationships between city-level gig hiring intensity and rural household outcomes. The horizontal axis is the prefecture-level city gig hiring intensity index, measured as $\ln(x+1)$. The four panels report household agricultural participation, transfer-out participation, transfer-out area, and total transfer area, with the latter two measured as $\ln(x+1)$. Solid lines show linear fits. The patterns are consistent with the baseline results: higher gig hiring intensity is associated with lower agricultural participation and greater land-transfer activity. Standard errors are clustered at the city level.

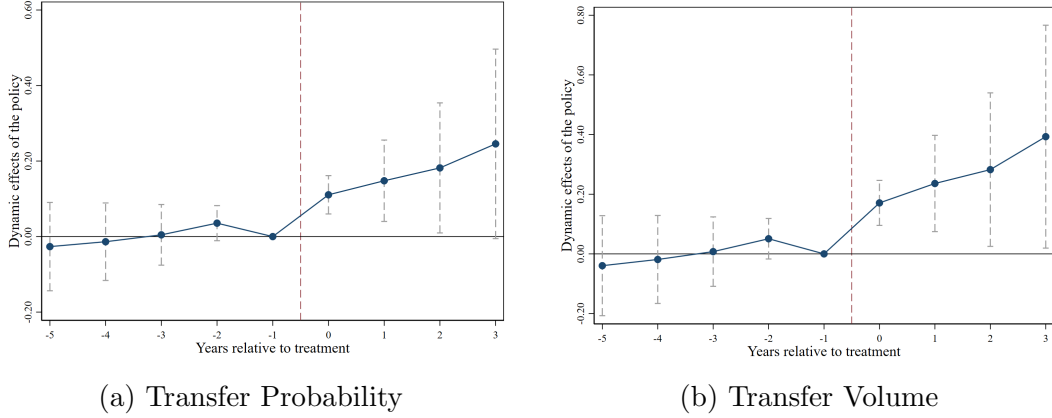


Figure 7. Dynamic Effects of Gig-Platform Entry: Event-Study Estimates of Rural Land Transformation

Notes: This figure plots event-study estimates of the effect of gig-platform entry on land transfer outcomes. The markers and capped spikes represent point estimates and 95 percent confidence intervals. The dependent variable in Panel (a) is an indicator equal to one if a household transfers land out. The dependent variable in Panel (b) is $\text{Log}(\text{Transfer-Out Area}+1)$, the natural logarithm of the land area transferred out plus one. The year immediately prior to platform entry is omitted as the reference period. All specifications incorporate household fixed effects, year fixed effects, village-specific linear time trends, and city-level controls.

ROBUSTNESS CHECKS

The main identification concern for the land-transfer results is that platform entry may not have been random across cities. If entry timing is correlated with underlying determinants of land transfer, the baseline estimates may partly capture selection rather than the effect of platform entry itself. We address this concern first and then consider a broader set of robustness checks, including spurious correlation, treatment-effect heterogeneity in staggered-adoption settings, sample composition, measurement error in treatment timing, household-specific trends, and alternative estimators and outcome transformations. To keep the main text concise, Table 4 reports the coefficient of interest for our two primary outcomes: transfer-out participation in Panel A and $\ln(\text{Transfer-Out Area} + 1)$ in Panel B.

Non-Random Platform Entry. A primary threat to identification is that platforms may have entered cities that were already on different trajectories of agricultural decline or land reallocation. Table 5 addresses this concern by regressing the platform entry year on pre-treatment slopes of household agricultural participation and land-transfer outcomes. Conditional on baseline city characteristics, these pre-treatment trends do not significantly predict entry timing. Column 1 of Table 4 further applies entropy-balancing weights to improve comparability between treated and untreated cities. The estimated effects remain positive for both primary outcomes and are similar in magnitude to the baseline estimates.¹²

¹²The corresponding full regression results for the entropy-balanced sample, together with analogous esti-

Spurious Correlation. A related concern is that the baseline estimates may reflect unobserved shocks rather than platform entry. We address this concern with a permutation exercise that randomly reassigns platform entry timing across cities 500 times. Figure 8 shows that the placebo estimates are centered near zero, while the actual estimate lies well outside the bulk of the placebo distribution. Panel (b) reports the bootstrap distribution of the baseline estimate and shows that the estimate is stable across resamples.

Treatment-Effect Heterogeneity. In staggered-adoption settings, the standard two-way fixed-effects estimator may be sensitive to heterogeneous treatment effects across cohorts or over time. We therefore re-estimate the dynamic effects using the interaction-weighted estimator of Callaway and Sant’Anna (2021). Figure 9 shows that the resulting group-time average treatment effects follow a pattern similar to the baseline event-study estimates.

Sample Composition and Treatment Timing. Columns 2 and 3 of Table 4 address two additional concerns. Column 2 excludes the four direct-administered municipalities, whose institutional and economic environments differ from those of other prefecture-level cities. Column 3 excludes cities for which the estimated platform entry year differs across the two cross-validation sources, reducing concern about measurement error in treatment timing. In both cases, the estimated effects remain positive and similar in magnitude to the baseline estimates.¹³

Household-Specific Trends. Column 4 of Table 4 adds household-specific linear time trends. The estimated effects remain positive and statistically significant for both primary outcomes, indicating that the baseline findings are not driven by differential household-level trends.

Functional Form. Columns 5 and 6 examine sensitivity to alternative estimators and outcome transformations. Column 5 reports Poisson pseudo-maximum-likelihood estimates, which are well suited to outcomes with many zeros. Column 6 replaces the log transformation with the inverse hyperbolic sine transformation for transfer-out area. The resulting estimates remain qualitatively similar to the baseline results.¹⁴

Overall, these exercises show that the positive effects of gig-platform entry on land transfer are robust to a wide range of alternative specifications and identification concerns.

V. Heterogeneity Analysis

Table 6 examines heterogeneity in the effect of gig-platform entry on agricultural participation by age and education. In Column 1, the coefficient on *Gig Platform Entry* $\times$ *Age* is positive and statistically significant, indicating that older individuals are less responsive to platform entry in their agricultural participation decisions. In Column 2, the coefficient on *Gig Platform Entry* $\times$ *Education* is negative and

mates for $\ln(\text{Total Transfer Area} + 1)$, are reported in Appendix Table A1.

¹³The corresponding full regression results for these restricted samples, together with analogous estimates for $\ln(\text{Total Transfer Area} + 1)$, are reported in Appendix Tables A3 and A4.

¹⁴Full results for these alternative specifications, together with analogous estimates for $\ln(\text{Total Transfer Area} + 1)$, are reported in Appendix Tables A6 and A5.

Table 4—Robustness Checks: The Impact of Gig-Platform Entry on Land Transfer

Variables	(1) EB Weights	(2) Excl. Muni.	(3) Cross-Validated	(4) HH Trends	(5) PPML	(6) IHS
Panel A: Land Transfer-Out Participation						
Gig Platform Entry	0.045*** (0.015)	0.023*** (0.008)	0.019** (0.009)	0.023*** (0.008)	0.117* (0.060)	— —
Observations	33,668	133,769	129,876	141,936	129,261	—
Adj./Pseudo R-squared	0.534	0.470	0.479	0.520	0.180	—
City Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	No	Yes
Village FE	No	No	No	No	Yes	No
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes	No	No	Yes
Household Linear Trends	No	No	No	Yes	No	No
Panel B: Log(Transfer-Out Area+1)						
Gig Platform Entry	0.062** (0.024)	0.034*** (0.012)	0.029** (0.013)	0.034*** (0.012)	0.139** (0.065)	0.041*** (0.015)
Observations	33,668	133,769	129,876	141,936	129,261	141,936
Adj./Pseudo R-squared	0.562	0.490	0.497	0.550	0.250	0.490
City Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	No	Yes
Village FE	No	No	No	No	Yes	No
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes	No	No	Yes
Household Linear Trends	No	No	No	Yes	No	No

Notes: This table reports a range of robustness checks for the baseline land-transfer results. Panel A uses an indicator for transfer-out participation, and Panel B uses transfer-out area, $\ln(x+1)$. Column 1 applies entropy-balancing weights based on baseline characteristics. Column 2 excludes the four direct-administered municipalities. Column 3 restricts the sample to cities with platform entry dates consistently verified across historical sources. Column 4 includes household-specific linear time trends. Column 5 reports Poisson pseudo-maximum-likelihood estimates. Column 6 replaces $\ln(x+1)$ with the inverse hyperbolic sine transformation for the continuous outcome. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5—Robustness: Predictability of Platform Entry Timing Using Pre-Treatment Trends

Variables	Platform Entry Year (Dependent Variable)			
	(1)	(2)	(3)	(4)
Pre-trend in Household Agricultural Participation	1.867 (1.252)	1.048 (1.075)	1.128 (1.044)	
Pre-trend in Land Transfer-Out Participation	0.830 (3.964)	3.413 (3.176)	−0.523 (1.056)	
Pre-trend in Log(Transfer-Out Area+1)	−0.697 (3.106)	−2.355 (2.611)		−0.251 (1.653)
Pre-trend in Log(Total Transfer Area+1)	−1.093 (1.632)	−0.687 (1.542)		−0.459 (1.611)
City-level Controls	No	Yes	Yes	Yes
Observations	207	207	207	207
R-squared	0.014	0.274	0.270	0.269

Notes: This table examines whether gig-platform entry timing is predicted by pre-treatment city trends. The dependent variable is the platform entry year. For each city, the pre-treatment slope is estimated by regressing the corresponding outcome on a linear time trend over the five years preceding platform entry, from $t - 5$ to $t - 1$. City-level controls are measured at their 2013 baseline values and include log population size, log GDP per capita, industrial structure, urbanization rate, and log mobile penetration. Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

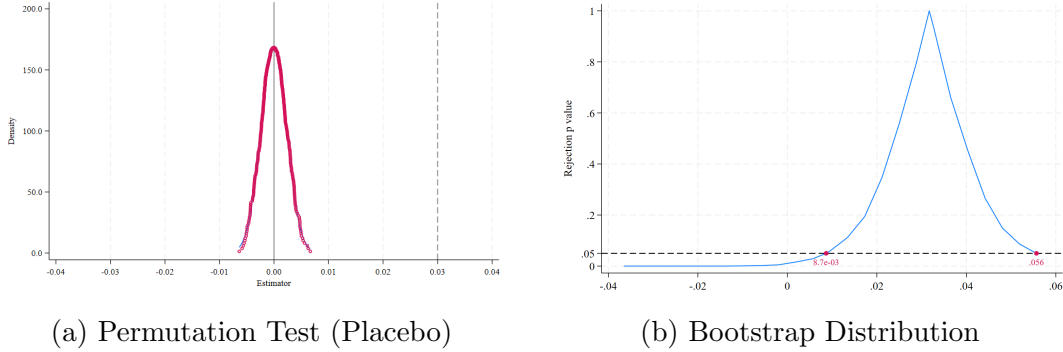


Figure 8. Falsification and Stability: Permutation Test and Bootstrap Distribution

Notes: This figure provides additional evidence on the robustness of the baseline estimates. Panel (a) presents the results of a permutation test based on 500 random assignments of gig-platform entry timing across cities. The curve represents the kernel density of the 500 placebo coefficients. The vertical dashed line indicates the actual baseline estimate. Panel (b) plots the distribution of estimates from a bootstrap procedure. Standard errors are clustered at the city level.

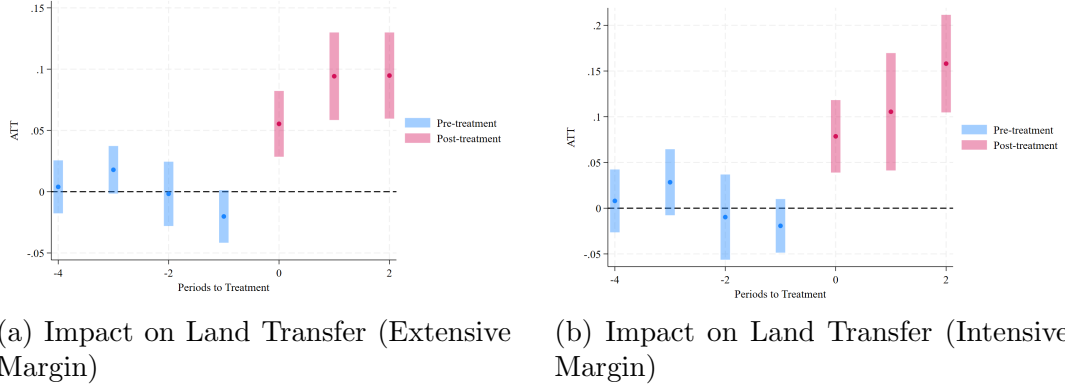


Figure 9. Robustness: Heterogeneous Treatment Effects Using the CS-DiD Estimator

Notes: This figure plots the dynamic treatment effects of gig-platform entry estimated using the doubly robust difference-in-differences estimator of Callaway and Sant’Anna (2021). Panel (a) reports the group-time average treatment effects on the treated (ATT) for land transfer-out participation, while Panel (b) reports the ATT for $\text{Log}(\text{Transfer-Out Area}+1)$. The solid dots represent point estimates, and the vertical lines denote 95 percent simultaneous confidence intervals. Standard errors are clustered at the city level.

statistically significant, indicating that the decline in agricultural participation is more pronounced for individuals with more years of schooling.

We next examine heterogeneity at the household level, focusing on whether labor endowments and land-market frictions shape the reallocation of labor and land following gig-platform entry. Table 7 considers one important source of such frictions: land fragmentation. We measure fragmentation using *Land Plots*, the total number of distinct plots in the household’s operational portfolio, which proxies for the transaction costs of land consolidation and leasing. The estimates indicate that households cultivating more fragmented land respond less strongly to platform entry. In Column 1, the reduction in agricultural participation associated with gig-platform entry is smaller for households with more land plots. In Columns 2 and 3, the interaction between *Gig Platform Entry* and *Land Plots* is negative and statistically significant, indicating that fragmentation weakens transfer-out responses on both the extensive and intensive margins. Across all three columns, the interaction term is significant at the 1 percent level. These findings suggest that, although the expansion of the gig economy promotes rural factor reallocation, the extent of adjustment remains constrained by pre-existing land fragmentation and the transaction costs it creates in land rental markets.

Table 8 further examines heterogeneity with respect to household labor endowments, proxied by *Labor Share*—the fraction of household members aged 15–70, used as a proxy for the internal labor endowment. Column 1 shows that the coefficient on *Gig Platform Entry* $\times$ *Labor Share* is positive and highly significant, indicating that the platform-induced decline in agricultural participation is substantially attenuated for households with an abundant supply of working-age labor. The interaction terms in Columns 2 and 3—*Gig Platform Entry* $\times$ *Labor Share* on land transfer-out

Table 6—Heterogeneity by Age and Education: Agricultural Participation

Variables	Individual Agricultural Participation		
	(1)	(2)	(3)
Gig Platform Entry	−0.0126*** (0.0035)	−0.0157*** (0.0035)	−0.0167*** (0.0036)
Gig Platform Entry × Age	0.0023*** (0.0003)		0.0017*** (0.0003)
Gig Platform Entry × Education		−0.0095*** (0.0011)	−0.0052*** (0.0009)
City-level Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes
Observations	355,919	355,919	355,919
Adj. R-squared	0.362	0.362	0.363

Notes: This table reports heterogeneity in agricultural participation responses. The sample is restricted to working-age individuals aged 15–70 (N=355,919), rather than the household-level sample used in Section 4 (N=141,936). The dependent variable is an indicator equal to one if the individual engages in agricultural employment. Heterogeneity is examined through interactions between gig-platform entry and individual characteristics, including age and years of schooling. Age is centered at its sample mean, and years of schooling is normalized so that zero corresponds to nine years of compulsory education; accordingly, the coefficient on *Gig Platform Entry* is interpreted for an individual of average age with nine years of schooling. All specifications include time-varying city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level.

Table 7—Household Heterogeneity: Land Fragmentation and Factor Reallocation

Variables	Household Agricultural Participation	Land Transfer-Out Participation	Log(Transfer-Out Area+1)
	(1)	(2)	(3)
Gig Platform Entry	−0.0262*** (0.0053)	0.0441*** (0.0088)	0.0695*** (0.0138)
Gig Platform Entry × Land Plots	0.0042*** (0.0007)	−0.0067*** (0.0012)	−0.0115*** (0.0022)
City-level Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes
Observations	141,936	141,936	141,936
Adj. R-squared	0.627	0.473	0.494

Notes: This table reports heterogeneous effects of gig-platform entry by household land fragmentation. *Land Plots*, defined as the total number of distinct plots in the household’s operational portfolio, proxies for land fragmentation and related transaction costs. The dependent variables are household agricultural participation, transfer-out participation, and transfer-out area, with the last measured as $\ln(x+1)$. All specifications include the interaction between *Gig Platform Entry* and *Land Plots*. All specifications also include time-varying city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

participation and $\text{Log}(\text{Transfer-Out Area}+1)$, respectively—are small in magnitude and statistically indistinguishable from zero, suggesting that the land-reallocation response to gig-platform entry is not mediated by household labor endowment. This suggests that while internal labor constraints dictate the labor allocation response, platform entry induces a broad-based increase in land leasing across households, irrespective of their initial demographic structure.

Table 8—Household Heterogeneity: Labor Endowment and Factor Reallocation

Variables	Household Agricultural Participation	Land Transfer-Out Participation	$\text{Log}(\text{Transfer-Out Area}+1)$
	(1)	(2)	(3)
Gig Platform Entry	−0.0821*** (0.0085)	0.0220** (0.0091)	0.0339** (0.0130)
Gig Platform Entry × Labor Share	0.1058*** (0.0100)	−0.0007 (0.0076)	−0.0044 (0.0117)
City-level Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes
Observations	141,321	141,321	141,321
Adj. R-squared	0.629	0.473	0.493

Notes: This table reports heterogeneous effects of platform entry by household labor endowment. *Labor Share*, defined as the share of household members aged 15–70, proxies for household labor endowment. All specifications include time-varying city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VI. Further Analysis

This section examines the broader economic adjustments induced by gig-platform entry. We proceed in two steps. We first study how platform entry affects household income and the allocation of employment across local and migrant labor markets. These results speak directly to the proposed mechanism that gig-platform expansion raises the relative return to local nonfarm work. We then examine how agricultural production adjusts following the induced reallocation of labor and land, focusing on output, revenue, and input use.

A. The Impact of Gig Economy on Income and Employment Allocation

We begin by examining whether gig-platform entry changes household income and the spatial allocation of labor. If gig-economy expansion induces land reallocation by improving local nonfarm earning opportunities, one would expect to observe changes in both household income composition and employment location. Table 9 reports the corresponding estimates. Column 1 shows that platform entry is associated with a 4.0-percent increase in log total household income (significant at the 10 percent level), consistent with the income-generating role of local gig opportunities. Looking across income sources and employment margins, the clearest effect appears in local

nonfarm earnings: Column 3 shows that platform entry raises log local nonfarm wage income by 0.220 log points (approximately 22.0 percent, significant at the 1 percent level). Column 2 indicates a 1.9-percentage-point increase in the local work share—the fraction of household members aged 15–70 with any local employment record—significant at the 10 percent level. By contrast, Columns 4 and 5 show no statistically significant effect on the out-migration work share or log migrant nonfarm wage income.

Overall, this pattern is consistent with a mechanism in which gig-platform expansion raises the relative return to local nonfarm employment, thereby increasing the opportunity cost of farm labor. This interpretation motivates the next question: how does agricultural production adjust when rural households reallocate labor away from farming and toward local nonfarm activities?

B. Agricultural Consequences of Gig-Economy Expansion

We next examine how agricultural outcomes respond to gig-platform entry. Table 10 reports the effects on production, revenue, and input use for the full sample, land-transferring-in households, and non-transferee households that remain in farming.

Panel A reports results for the full sample. Column 1 shows that platform entry reduces log food production by 0.1717 log points, significant at the 10 percent level. Columns 2 through 6 show no statistically significant effects on food yield per mu, crop revenue per mu, food revenue per mu, cash revenue per mu, or capital per mu. The full-sample results therefore point to a decline in overall food production without corresponding changes in per-mu revenue or capital intensity among households that remain in agricultural production.

Panel B restricts the sample to households that transfer in land. Several differences emerge for this group. Column 6 shows that platform entry increases log capital per mu by 0.3084 log points, significant at the 1 percent level. Column 3 shows that crop revenue per mu rises by 0.1763 log points, significant at the 10 percent level. At the same time, Column 2 shows that food yield per mu declines by 0.1301 log points, significant at the 1 percent level. The coefficients on food production, food revenue per mu, and cash revenue per mu are not statistically significant.

The estimates indicate that land-transferring-in households respond to platform entry by increasing capital input and achieving higher crop revenue per mu, but not higher food output or food yield. The decline in food yield per mu suggests that the adjustment does not take the form of broad-based productivity gains in food production.

Panel C restricts the sample to households that do not transfer in land but remain engaged in farming. For this group, the coefficients on food production and food yield per mu are statistically insignificant. Crop revenue per mu declines by 0.0640 log points, significant at the 10 percent level, and cash revenue per mu declines by 0.0580 log points, also significant at the 10 percent level. The coefficients on food revenue per mu and capital per mu are not statistically significant.

The results suggest that non-transferee households do not experience improvements in agricultural performance following platform entry. If anything, their per-mu rev-

enue declines modestly.

C. Summary

Taken together, the further analysis shows that gig-platform entry affects rural households through changes in both labor allocation and agricultural outcomes. On the labor side, platform entry raises local nonfarm employment and local nonfarm wage income, but does not affect migrant employment or migrant wage income. On the agricultural side, platform entry reduces food production in the full sample and generates heterogeneous effects across farming households. Among land-transferring-in households, it raises capital input per mu and crop revenue per mu while lowering food yield per mu. Among non-transferee households, it lowers crop revenue per mu. The broader pattern indicates that gig-platform expansion facilitates reallocation out of agriculture, but its agricultural consequences are uneven across households.

Table 9—Further Analysis: The Impact of Platform Entry on Income and Employment Allocation

Variables	Total	Local		Out-Migration	
	Income (1)	Work Share (2)	Nonfarm Income (3)	Work Share (4)	Nonfarm Income (5)
Gig Platform Entry	0.040* (0.021)	0.019* (0.011)	0.220*** (0.085)	−0.012 (0.011)	−0.099 (0.091)
City Controls	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes	Yes	Yes
Observations	141,936	130,073	141,936	130,073	141,935
Adj. R-squared	0.602	0.529	0.530	0.621	0.590

Notes: This table reports the effects of gig platform entry on rural household income and employment allocation. The dependent variables are total household income, local work share, local nonfarm wage income, out-migration work share, and migrant nonfarm wage income. Income variables are measured as $\ln(x + 1)$. The sample sizes in Columns 2 and 4 are naturally smaller, as households with zero working-age members are excluded from the labor share calculation. All specifications incorporate time-varying city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VII. Conclusion

This paper examines how the expansion of the gig economy affects rural factor allocation in China. Using city-level platform entry dates constructed from historical web archives from the Wayback Machine and matching them to household panel data, we estimate a staggered difference-in-differences model.

We find that gig-platform entry reduces household agricultural participation and increases land transfer-out on both the extensive and intensive margins. Platform entry also raises local nonfarm employment and local nonfarm wage income, but has no statistically significant effect on out-migration employment or migrant nonfarm

Table 10—Further Analysis: Agricultural Consequences by Land Transfer Status

Variables	Production		Revenue (per Mu)			Capital
	Food Production (1)	Food Yield (2)	Total (3)	Food (4)	Cash (5)	per Mu (6)
Panel A: Full Sample						
Gig Platform Entry	−0.1717* (0.0971)	−0.0121 (0.0260)	−0.0108 (0.0725)	−0.0196 (0.0335)	−0.0491 (0.0340)	−0.0329 (0.0510)
Observations	141,936	83,762	141,936	83,761	57,017	141,936
Adj. R-squared	0.775	0.519	0.743	0.483	0.656	0.769
City Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: Land-Transferring-In Households						
Gig Platform Entry	−0.0245 (0.0476)	−0.1301*** (0.0492)	0.1763* (0.0913)	−0.0540 (0.0582)	−0.0578 (0.1260)	0.3084*** (0.1168)
Observations	4,811	4,811	6,116	4,811	3,650	6,116
Adj. R-squared	0.853	0.657	0.527	0.521	0.695	0.732
City Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes	Yes	Yes	Yes
Panel C: Non-Transferee Households						
Gig Platform Entry	−0.0900 (0.0880)	−0.0180 (0.0280)	−0.0640* (0.0380)	−0.0170 (0.0340)	−0.0580* (0.0340)	−0.0410 (0.0460)
Observations	89,328	72,922	89,328	72,921	47,958	89,328
Adj. R-squared	0.818	0.537	0.602	0.496	0.663	0.789
City Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports heterogeneous effects of gig-platform entry on agricultural production, revenue, and input use. The dependent variables are food production, food yield per mu, crop revenue per mu, food revenue per mu, cash revenue per mu, and productive fixed assets per mu. All dependent variables are measured as $\log(x + 1)$. Panel A reports results for the full sample. Panel B restricts the sample to land-transferring-in households. Panel C restricts the sample to non-transferee households that remained actively engaged in farming. All specifications include city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

wage income. The effects are stronger among younger and more educated individuals, while land fragmentation significantly constrains land reallocation.

On the agricultural side, the effects are heterogeneous across households. In the full sample, platform entry reduces food production. Among land-transferring-in households, it increases capital input per mu and crop revenue per mu while reducing food yield per mu. Among non-transferee households, it lowers crop revenue per mu. These results suggest that the expansion of the gig economy can contribute to reallocation out of agriculture through local labor-market adjustment, without relying on large-scale migration. At the same time, the agricultural consequences of this process are uneven, and the observed reallocation is not associated with broad-based improvements in agricultural performance.

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APPENDIX



Figure A1. Word Cloud of Gig Economy Occupations

Notes: This figure presents a word cloud based on the self-reported job descriptions of gig workers in the CLDS sample. Because the survey uses the traditional term “*lingsan yonggong*” rather than the contemporary concept of the gig economy, we use respondents’ occupational descriptions to construct the figure. Word size is proportional to frequency in the survey responses.

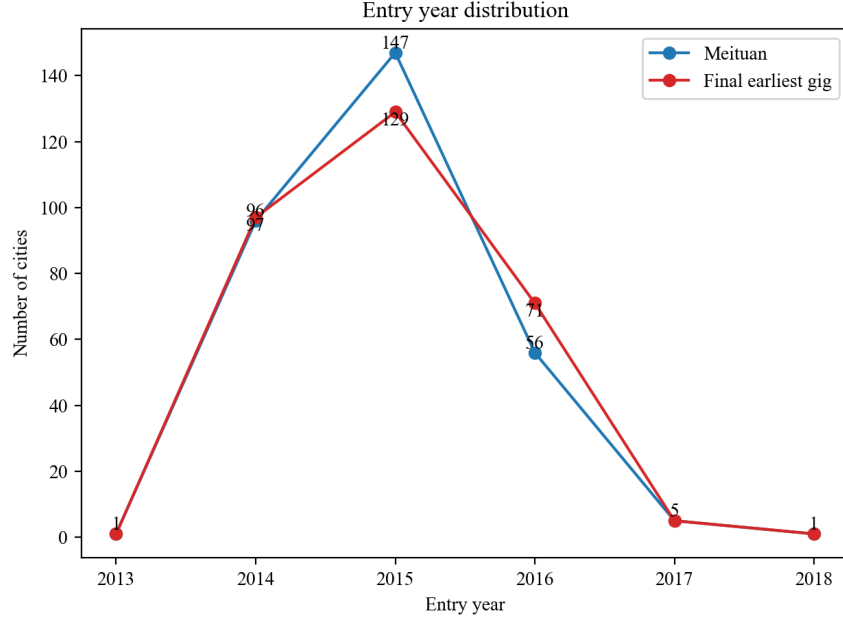


Figure A2. Validation of Gig-Platform Entry Dates: Meituan Entry Years and Final Adopted Entry Years

Notes: This figure cross-validates our platform entry dates against the expansion records of Meituan. The entry dates are constructed from historical platform web archives and cross-validated with Zhilian Recruitment data. The figure plots the distribution of the earliest platform entry year across cities. The overlap between the two series supports the reliability of the treatment timing.

Table A1—Robustness to Entropy-Balancing Weights

Variables	Land Transfer-Out Participation	Log(Transfer-Out Area+1)	Log(Total Transfer Area+1)
	(1)	(2)	(3)
Gig Platform Entry	0.045*** (0.015)	0.062** (0.024)	0.061 (0.044)
City-level Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes
Observations	33,668	33,668	33,668
Adj. R-squared	0.534	0.562	0.549

Notes: This table presents entropy-balanced estimates addressing non-random platform entry. Column 1 uses an indicator for land transfer-out participation. Column 2 reports estimates for Log(Transfer-Out Area+1). Column 3 reports estimates for Log(Total Transfer Area+1). All specifications incorporate time-varying city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A2—Robustness: Controlling for Household-Specific Linear Time Trends

Variables	Land Transfer-Out Participation	Log(Transfer-Out Area+1)	Log(Total Transfer Area+1)
	(1)	(2)	(3)
Gig Platform Entry	0.023*** (0.008)	0.034*** (0.012)	0.042*** (0.015)
City-level Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Household Linear Trends	Yes	Yes	Yes
Observations	141,936	141,936	141,936
Adj. R-squared	0.52	0.55	0.56

Notes: This table presents robustness estimates incorporating household-specific linear time trends. All specifications incorporate time-varying city-level controls, household fixed effects, year fixed effects, and household-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3—Robustness: Excluding Direct-Administered Municipalities

Variables	Land Transfer-Out Participation	Log(Transfer-Out Area+1)	Log(Total Transfer Area+1)
	(1)	(2)	(3)
Gig Platform Entry	0.023*** (0.008)	0.034*** (0.012)	0.043*** (0.014)
City-level Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes
Observations	133,769	133,769	133,769
Adj. R-squared	0.47	0.49	0.50

Notes: This table presents estimates excluding observations from Beijing, Tianjin, Shanghai, and Chongqing. Column 1 uses an indicator for land transfer-out participation. Column 2 reports estimates for Log(Transfer-Out Area+1). Column 3 reports estimates for Log(Total Transfer Area+1). All specifications incorporate time-varying city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4—Robustness: Excluding Cities with Non-Cross-Validated Entry Dates

Variables	Land Transfer-Out Participation	Log(Transfer-Out Area+1)	Log(Total Transfer Area+1)
	(1)	(2)	(3)
Gig Platform Entry	0.019** (0.009)	0.029** (0.013)	0.035** (0.015)
City-level Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes
Observations	129,876	129,876	129,876
Adj. R-squared	0.479	0.497	0.493

Notes: This table excludes 22 prefecture-level cities whose platform entry dates cannot be cross-validated across the two historical sources. The dependent variables are an indicator for transfer-out participation, transfer-out area, and total transfer area, with the latter two measured as $\ln(x+1)$. All specifications include time-varying city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5—Robustness: Inverse Hyperbolic Sine (IHS) Transformation

Variables	IHS(Transfer-Out Area)	IHS(Total Transfer Area)
	(1)	(2)
Gig Platform Entry	0.041*** (0.015)	0.051*** (0.017)
City-level Controls	Yes	Yes
Household FE	Yes	Yes
Year FE	Yes	Yes
Village Linear Trends	Yes	Yes
Observations	141,936	141,936
Adj. R-squared	0.49	0.50

Notes: This table reports baseline estimates using the inverse hyperbolic sine transformation for the dependent variables. Column 1 uses IHS(Transfer-Out Area), and Column 2 uses IHS(Total Transfer Area). All specifications include time-varying city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6—Robustness to PPML Estimator with Village Fixed Effects

Variables	Land Transfer-Out Participation	Transfer-Out Area (Level)	Total Transfer Area (Level)
	(1)	(2)	(3)
Gig Platform Entry	0.117* (0.060)	0.139** (0.065)	0.126** (0.053)
City-level Controls	Yes	Yes	Yes
Village FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	129,261	129,261	131,343
Pseudo R-squared	0.18	0.25	0.33

Notes: This table reports estimates using the Poisson pseudo-maximum-likelihood estimator. Column 1 uses an indicator for transfer-out participation, Column 2 uses transfer-out area in levels, and Column 3 uses total transfer area in levels. The coefficient in Column 1 is interpreted as a semi-elasticity of the conditional expectation rather than a marginal probability effect. All specifications include time-varying city-level controls, village fixed effects, and year fixed effects. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A7—Further Analysis: Agricultural Consequences in the Full Sample

Variables	Production		Revenue (per Mu)			Capital
	Food Production (1)	Food Yield (2)	Total (3)	Food (4)	Cash (5)	per Mu (6)
Gig Platform Entry	-0.1717* (0.0971)	-0.0121 (0.0260)	-0.0108 (0.0725)	-0.0196 (0.0335)	-0.0491 (0.0340)	-0.0329 (0.0510)
City Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes	Yes	Yes	Yes
Observations	141,936	83,762	141,936	83,761	57,017	141,936
Adj. R-squared	0.775	0.519	0.743	0.483	0.656	0.769

Notes: This table reports the effects of gig-platform entry on agricultural production, revenue, and input use for the full sample. Dependent variables: Log(Food Production+1) (Column 1); Log(Food Yield per Mu+1) (Column 2); Log(Crop Revenue per Mu+1) (Column 3); Log(Food Revenue per Mu+1) (Column 4); Log(Cash Revenue per Mu+1) (Column 5); Log(Capital per Mu+1) (Column 6). Sample sizes differ across columns because Columns 1, 2, and 4 condition on positive food-crop sown area, and Column 5 conditions on positive cash-crop sown area. All specifications incorporate city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8—Further Analysis: Agricultural Consequences among Land-Transferring-In Households

Variables	Production		Revenue (per Mu)			Capital
	Food Production (1)	Food Yield (2)	Total (3)	Food (4)	Cash (5)	per Mu (6)
Gig Platform Entry	−0.0245 (0.0476)	−0.1301*** (0.0492)	0.1763* (0.0913)	−0.0540 (0.0582)	−0.0578 (0.1260)	0.3084*** (0.1168)
City Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,811	4,811	6,116	4,811	3,650	6,116
Adj. R-squared	0.853	0.657	0.527	0.521	0.695	0.732

Notes: This table reports the effects of gig-platform entry on agricultural production, revenue, and input use for the subsample of land-transferring-in households ($area_{in} > 0$). Dependent variables: Log(Food Production+1) (Column 1); Log(Food Yield per Mu+1) (Column 2); Log(Crop Revenue per Mu+1) (Column 3); Log(Food Revenue per Mu+1) (Column 4); Log(Cash Revenue per Mu+1) (Column 5); Log(Capital per Mu+1) (Column 6). Sample sizes differ across columns because Columns 1, 2, and 4 condition on positive food-crop sown area, and Column 5 conditions on positive cash-crop sown area. All specifications include city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A9—Further Analysis: Agricultural Consequences among Non-Transferee Households

Variables	Production		Revenue (per Mu)			Capital
	Food Production (1)	Food Yield (2)	Total (3)	Food (4)	Cash (5)	per Mu (6)
Gig Platform Entry	−0.090 (0.088)	−0.018 (0.028)	−0.064* (0.038)	−0.017 (0.034)	−0.058* (0.034)	−0.041 (0.046)
City Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Village Linear Trends	Yes	Yes	Yes	Yes	Yes	Yes
Observations	89,328	72,922	89,328	72,921	47,958	89,328
Adj. R-squared	0.818	0.537	0.602	0.496	0.663	0.789

Notes: This table reports the effects of gig-platform entry on agricultural production, revenue, and input use for the subsample of non-transferee households ($area_{in} = 0$) with active farming operations ($area_e > 0$) and agricultural labor. Dependent variables: Log(Food Production+1) (Column 1); Log(Food Yield per Mu+1) (Column 2); Log(Crop Revenue per Mu+1) (Column 3); Log(Food Revenue per Mu+1) (Column 4); Log(Cash Revenue per Mu+1) (Column 5); Log(Capital per Mu+1) (Column 6). Sample sizes differ across columns because Columns 1, 2, and 4 condition on positive food-crop sown area, and Column 5 conditions on positive cash-crop sown area. All specifications include city-level controls, household fixed effects, year fixed effects, and village-specific linear time trends. Standard errors are clustered at the city level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.